Gregory Aiosa, EI

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Education

Northwestern University, Master of Science in Robotics **Sept 2025 – Dec 2026**

- **Coursework:** Embedded Systems in Robotics, Robotic Manipulation, Intro to Mechatronics

University of Florida, Bachelor of Science in Mechanical Engineering **Aug 2016 – Dec 2020**

- **Honors:** Cum Laude
- **Coursework:** Control of Mechanical Engineering Systems, Geometry of Robots and Mechanisms

Experience

Mechanical Engineer, Slice Engineering – Gainesville, FL **Jan 2021 – Jul 2025**

- Lead design engineer for industrial 3D printer end-effector, ensuring compliance with MIL-STD 810H and ITAR
- Utilized GD&T principles to prepare engineering drawings sent to contract manufacturers to create hotends
- Devised test procedures to optimize polymer melt, driving a 70% increase in volumetric flow rate via iteration
- Developed Python scripts to process thermal sensor data, automating the validation of mechatronic end-effectors
- Conducted FAIs and established inspection protocols to prevent defects and capture data for future iterations

Projects

7-DOF Robot Domino Artist **Fall 2025**

- Collaborated with a team to develop a Python ROS 2 package for a Franka Panda arm to find and manipulate dominoes
- Implemented motion planning logic to manipulate dominoes while avoiding collision objects defined in the scene
- Utilized OpenCV, AprilTags, and a RealSense camera for domino detection and extrinsic calibration

Pen Grasping Robot **Fall 2025**

- Programmed the Interbotix PincherX-100 4-DOF robot arm to grab a pen located via a RealSense camera using OpenCV

Bio 3D Printer **Fall 2019 – Spring 2020**

- Collaborated with a team of 4 to design the motion system for a delta-style bio 3D printer weighing less than 100 grams
- Modeled custom aluminum joints using SolidWorks and sourced lightweight, low-friction rod end bearings
- Led the electro-mechanical integration of the motion system with three subsystems, managing interface tolerances

PID Control of Floating Ping Pong Ball **Fall 2020**

- Moved a ping pong ball to 3 positions in a tube using a fan while minimizing steady-state error to 5 mm for 3 seconds
- Computed data from a TOF sensor into a DAQ with LabVIEW and output a PID control signal to the motor controller

Combat Robots **Fall 2017 – Spring 2020**

- Built a 3 lb combat robot with a drum-style weapon powered by a 13,000 RPM brushless DC motor
- Designed a 15 lb combat robot in SolidWorks, featuring an edger blade powered by reused hammer drill motors

Skills

Robotics: ROS 2, Control Systems, Embedded Systems, MoveIt, OpenCV, Gazebo, RViz

Programming Languages: Python, MATLAB

Software: SolidWorks (CSWA)/Onshape/Fusion360, Git, Linux, CoppeliaSim, LabVIEW

Hardware: GD&T, Arduino, Raspberry Pi, 3D Printing, Soldering, Manual Machining, Infrared Thermography